Unit 5: Understand the Data

Student Checklist · Data Science and AI EPQ

Name: _____ **Date started:** _____

How to use this checklist

This sheet is optional. Everything you need is on the unit page. Tick each item as you finish it so you can find your place again. The unit should take about **3 hours**.

Write your own answer before you reveal the model answer. Then compare the two and note anything you missed.

Before you start: have the Orange workflow you saved at the end of Unit 4 ready. If you no longer have it, download `health_data.csv` from the link in Activity 5.1.

Keep in view

Research question: "Can health and lifestyle data be used to estimate an individual's Health Score?"

Target variable: `Health_Score`, the value you are trying to predict.

Already found in Unit 4: 50 `Person_ID`s appear more than once · the UK is recorded as both 'UK' and 'United Kingdom' in `Country` · the sample is not fully representative (about 40% gym members, 4% non-binary).

1. Recap of Unit 4

☐ I have read the recap. The three Unit 4 issues are already in the data issues log (Section 5).

2. Understanding the Structure of the Dataset

☐ I can explain the difference between an **observation** (a row) and a **variable** (a column).

☐ I can explain what a **target variable** is, and name the target in this investigation.

☐ I can identify variable types: numerical (continuous, discrete) and categorical (nominal, ordinal, binary).

Activity 5.1 – The Structure of the Dataset

☐ Opened my Unit 4 workflow and deleted the **Distributions**, **Group By** and second **Data Table** widgets.

☐ Opened the **Data Table** and looked across the rows and columns.

Questions ☐ (a) ☐ (b) ☐ (c) ☐ (d)

Tip for 5.1 (d): this is a long table. Group the variables by theme (about the person, sleep, exercise, diet…) to keep track.

3. Exploring the Variables

☐ I can describe a categorical variable using its categories, frequency, proportion and **mode**.

☐ I can describe a numerical variable using **min, max, range, mean and median**, and say when the median is better than the mean.

Activity 5.2 – Exploring Two Variables (Column Statistics)

☐ Added a **Column Statistics** widget and found Marital_Status and Daily_Steps.

Questions ☐ (a) ☐ (b)

☐ I can explain what a **missing value** is, and why patterns in where values are missing matter.

Activity 5.3 – Missing Values

☐ Part 1: sorted the **Missing** column in Column Statistics.

Question ☐ (a)

☐ Part 2: sorted the **Data Table** by BMI, Exercise_Type and Wake_Up_Time and read across the rows.

Questions ☐ (b) ☐ (c) ☐ (d)

☐ I can explain the difference between an **outlier** and an **anomaly**, and why the terms are not always used consistently.

☐ I can recognise a **right-skewed** and a **left-skewed** distribution, and say how skew affects the mean.

Activity 5.4 – Unusual Values (Distributions)

☐ Part 1: added a **Distributions** widget and checked BMI (bin width 10), Working_Hours_Per_Day and Outdoor_Time_Hours.

Questions ☐ (a) ☐ (b) ☐ (c)

☐ Part 2: sorted the **Data Table** to look at the highest BMI values and both ends of Working_Hours_Per_Day.

Questions ☐ (d) ☐ (e)

☐ I have added every issue I found in Activities 5.3 and 5.4 to my data issues log (Section 5).

4. Identifying Patterns

☐ I can describe a Pearson correlation as **weak** (below 0.3), **moderate** (0.3–0.7) or **strong** (above 0.7), and explain that the sign only shows direction.

☐ I can explain **target leakage** and **multicollinearity** in my own words.

☐ I can give three limitations of correlation: it is distorted by wrong values, it only detects linear relationships, and it does not show cause.

Activity 5.5 – Correlations with Health_Score

☐ Added a **Correlations** widget and set Health_Score as the variable to correlate against.

Questions ☐ (a) ☐ (b) ☐ (c) ☐ (d)

Activity 5.6 – Checking a Correlation with a Scatter Plot

☐ Added a **Scatter Plot** with BMI on the X axis, Health_Score on the Y axis and the regression line shown.

Questions ☐ (a) ☐ (b)

☐ Looked at how some other variables relate to Health_Score in the scatter plot.

☐ **Saved my Orange workflow** for Unit 6.

5. Evaluating the Dataset

☐ I can explain why data scientists record problems **before** changing anything.

Activity 5.7 – Preparing for Unit 6

Use these tables as your two lists. Add issues as you find them in Sections 3 and 4. For each one, decide whether it can be **fixed from the data**, **cannot be recovered** (estimate or set aside), or is a **limitation to remember**.

Data issues log

Issue	Variable(s)	Found in	What needs to happen?
Duplicate records (50 Person_IDs)	Person_ID	Unit 4	
UK recorded under two names	Country	Unit 4	
Sample not fully representative	Gym_Member, Gender	Unit 4	

Variables to explore further, or treat with caution

Variable	Explore or caution?	Why? (correlation, type, possible leakage…)

Questions ☐ (a) ☐ (b) ☐ (c)

☐ Kept both lists safe to use in Unit 6.

6. Unit 5: Review

☐ Read the Summary of the Unit.

☐ Answered the five Reflection Questions. They are good to discuss with a partner or mentor.

☐ Optional: tried the Extension Activity (review the documentation for a public dataset).

How confident do I feel?

Tick each line you are confident about. If you are not sure yet, leave it empty and ask your teacher or mentor.

☐ I can describe how a dataset is organised into observations and variables.

☐ I can tell the target variable apart from the other variables.

☐ I can classify a variable as numerical or categorical.

☐ I can find and interpret the mean, median, mode and range of a numerical variable.

☐ I can describe the distribution of a categorical variable.

☐ I can identify outliers and missing values in a dataset.

☐ I can interpret a correlation coefficient and explain its limitations.

☐ I can judge which variables are likely to be useful, weak or unsuitable for prediction.

Ready for Unit 6? ☐ Orange workflow saved ☐ Both Activity 5.7 lists kept ☐ I know what I need help with